CSA4307 - Internet Programming

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CIA-I

1. Internet Protocols and HTTP communication

a) TCP/IP, DNS and URL

- TCP/IP : Helps transfer data between the browser and server reliably.
- DNS : Converts domains names into IP address.
- URL : Specifies the location of web resource

Ex: https://example.com/trains/search.

b) HTTP Request and Response

HTTP Request contains:-

1. Request line - Method and URL
2. Headers - Additional information
3. Body - Data sent to servers

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HTTP Response contains:-

1. Status line - Response status
2. Headers - Content information
3. Body - Returned data

c) HTTP Methods

- GET - Retrieve data, safe and Indempotent
- POST - Submit or create data,
- PUT - Update or replace data, it is Indempotent.
- DELETE - Delete Data, it is Indempotent.

d) HTTP Status codes

- 200 - Request successfull
- 301 - Permanently moved
- 304 - Not modified.
- 401 - Unathorized
- 404 - page not found
- 500 - Interval Server error.
- 503 - Server unabalable.

Problem & Solution:

- Duplicate booking :- Post submited again so use pattern/Idempotent key
- state seat details :- Browser cacheing use cache control.
- Unexpected logout :- cookies/sesion expiry improve sesion manage
- Data theift :- Insecure connection USE-HTTPS

* Secure cookies should use secure, HttpOnly, SameSite attributes
* conclution: Proper HTTP methods, caching, submitting, sessions and HTTPS improve the security and reliability of railway booking.

2. XHTML Cousse Registration Page:-

a) Document Structure:-

XHTML is an XML-based version of HTML that follows strict syntax rules

- Doctoon
- DOCTYPE - Define document type
- html - Root element.
- head - contains metadata.
- body - contains whole content.
- header. - Page loading
- meta - Defines page information.
- nav - Navigation links
- main - main content.
- section - Groups content.
- footer - Bottom sections.

b) Table elements

- table - creates table.
- caption - Table title
- thead - Table headings
- tbody - Table data
- th - Heading cell
- colspan - combines columns.
- Rowspan - combines rows
- Scope - Identifies heading direction

c) Form Elements

- form - Creates a form
- action - specifies where data sent
- method - specifies GET or POST
- label - Describes input fields
- input - Accepts user info
- select - Creates dropdowns
- textarea - Accepts comments
- pattern - validates input format

d) Hyperlink, Images and URLS

Hyperlink : Home .

Image :

Relative URL : Refers to file location.

Absolute URL : contains complete web address.

HTML:

- Flexible Syntax
- Closing tabs optional
- Tags are case insensitive
- Some attributes may be opted.
- Flexible nestings

XHTML:

- Strict syntax
- Every element must close
- Tags must be lowercase.
- Attributes must be quoted.
- Proper nesting required.

*POST is preffered for registration because students infor about should not be appear in URL parameters.

3. a) XML Library catalogue

Introduction :-

XML stands for extensible Markup Language. it is us to store and exchange structured data between app

Ex:- <?xml version = "1.0"?>
<catalogue xmlns = "http://example.com/library">
<book isbn = "101">
..... //book details.
</book>
</catalogue>

XML Schema (XSD): XSD defines the structure & data types of XML documents.

XML	HTML	JSON
Stores data	Displays webpages	Exchange Data.
custom tags	Predefined tags	key value format.
Strict syntax	Flexible syntax	Light weights.

XSLT converts XML data into HTML for display. CSS can be used to style XML elements.

b) CSS cascade and selectors

* From Given HTML and CSS final answer colour is "orange" because inline css has higher priority than normal ID, class and element selectors.
* If !important is added at green, then the final colours becomes GREEN. because an important author declaration overrides the normal inline

Types of CSS:

1. Inline CSS - Written inside an HTML element.
2. Internal CSS - written inside the <style> element.
3. External CSS - written in a seperate .css file.

* CSS cascad and selectors control webpage appearances and improve styling consistency.

Q4. Responsive Web design Using CSS:

Responsive webpage makes websites adjusts automatically to mobile phones, tablets and desktops.

a) CSS Box Model:

1. content - Actual information.
2. Padding - space around content
3. Border - surrounds content and padding.
4. Margin - space outside the border.

b) CSS positioning

- Static - Default position.
- Relative - Moves from normal position.
- Absolute - Positioned relative to an ancestor
- Fixed - Stays fixed on screen.
- Sticky - sticks during scrolling.

d) HTTP Status codes

c) flex box and Grid

Flexbox:-

- one - dimensional layout.
- Used for navigation bars.
- Aligns items in rows and columns.

Grid:-

- Two dimensional layout.
- Used for main content and sidebars.
- Creates rows and columns.

d) Responsive Techniques.

- %. - Relative width
- rem - Relative font sizing.
- vw - Viewpoint width.
- Viewport meta tag.
- Media quires
- Flexible images.

e) Typography, colours and css variables

- Use readable fonts and proper font sizes.
- Maintain sufficient contrast between text and backgrounds
- Use css variables for consistent colour and themes.

Interactive online Bookstore Using Javascripts

Java script makes webpages interactive. In an online bookstore, it helps update cart totals, remove books validate forms and retrieve info without refreshing pages.

a) Java script basics:

- let - variables whose values can change.
- const - variable that cannot be reassigned.
- Datatypes, Arrays, object, function, Arrow Function.

Ex:

```
let quantity = 2;
const price = 500;
const book = {
        name = "Java";
        price : 500;
};
const total = (price, qty) => price * qty;
```

b) DOM manipulation:

It represents the webpages as tree structure

- getElementById() - selects an element by ID.
- querySelector()
- createElement()
- textContent()
- classList().

c) Event handling:

Events responds to user actions.

* click
* input
* change
* submit

d) Regular Expression validation

Regular expression validates user inputs

* Name
* Email
* Phone number
* Password.

e) Fetch API and Asynchronous communication.

- Promise represents future results.
- Async/await makes asynchronous code easier to read.
- JSON is used to exchange structured data.
- Error handling manages faild requests

* Javascripts provides dynamic shopping-cart operations, form validation and asynchronous communication.